

Vector Analysis

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1 Vector Identities

- Triple Products
- Gradient Identities
- Divergence Identities
- Curl Identities
- Second Derivative Identities

2 Orthogonal Coordinate Systems

- Cartesian Coordinates (x, y, z)
 - Vector Operations
- Cylindrical Coordinates (ρ, ϕ, z)
 - Coordinate Transformation
 - Vector Operations
- Spherical Coordinates (r, θ, ϕ)
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Chapter Overview

This appendix summarizes only the essential vector identities and transformation formulas that are directly used in the derivations of the main text.

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Triple Products

The scalar triple product of three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ is given by:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}). \quad (\text{A.1.1})$$

This can also be written as the determinant of the matrix formed by the three vectors:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \quad (\text{A.1.2})$$

The vector triple product is given by the "BAC-CAB" rule:

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B}). \quad (\text{A.1.3})$$

Gradient Identities

For scalar fields f and g , and vector fields $\vec{A}$ and $\vec{B}$:

$$\vec{\nabla}(fg) = f \vec{\nabla}g + g \vec{\nabla}f, \quad (\text{A.1.4})$$

$$\begin{aligned} \vec{\nabla}(\vec{A} \cdot \vec{B}) &= (\vec{A} \cdot \vec{\nabla}) \vec{B} + (\vec{B} \cdot \vec{\nabla}) \vec{A} \\ &\quad + \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A}). \end{aligned} \quad (\text{A.1.5})$$

Divergence Identities

For a scalar field f and vector fields $\vec{A}$ and $\vec{B}$:

$$\vec{\nabla} \cdot (f\vec{A}) = f (\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla} f), \quad (\text{A.1.6})$$

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}). \quad (\text{A.1.7})$$

Curl Identities

For a scalar field f and vector fields $\vec{A}$ and $\vec{B}$:

$$\vec{\nabla} \times (f\vec{A}) = f(\vec{\nabla} \times \vec{A}) + (\vec{\nabla} f) \times \vec{A}, \quad (\text{A.1.8})$$

$$\begin{aligned} \vec{\nabla} \times (\vec{A} \times \vec{B}) &= \vec{A}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{A}) \\ &\quad + (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B}. \end{aligned} \quad (\text{A.1.9})$$

Second Derivative Identities

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0, \quad (\text{A.1.10})$$

$$\vec{\nabla} \times (\vec{\nabla} f) = 0, \quad (\text{A.1.11})$$

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f \quad (\text{Laplacian}), \quad (\text{A.1.12})$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}. \quad (\text{A.1.13})$$

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$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}, \quad (\text{A.2.1})$$

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}, \quad (\text{A.2.2})$$

$$\begin{aligned} \vec{\nabla} \times \vec{A} = & \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ & + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ & + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right), \end{aligned} \quad (\text{A.2.3})$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}. \quad (\text{A.2.4})$$

Coordinate Transformation

$$x = \rho \cos \phi, \quad (\text{A.2.5})$$

$$y = \rho \sin \phi. \quad (\text{A.2.6})$$

$$\vec{\nabla} f = \hat{\rho} \frac{\partial f}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial f}{\partial \phi} + \hat{z} \frac{\partial f}{\partial z}, \quad (\text{A.2.7})$$

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}, \quad (\text{A.2.8})$$

$$\begin{aligned} \vec{\nabla} \times \vec{A} = & \hat{\rho} \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \\ & + \hat{\phi} \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \\ & + \hat{z} \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right], \end{aligned} \quad (\text{A.2.9})$$

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}. \quad (\text{A.2.10})$$

Coordinate Transformation

$$x = r \sin \theta \cos \phi, \quad (\text{A.2.11})$$

$$y = r \sin \theta \sin \phi, \quad (\text{A.2.12})$$

$$z = r \cos \theta. \quad (\text{A.2.13})$$

$$\vec{\nabla} f = \hat{r} \frac{\partial f}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}, \quad (\text{A.2.14})$$

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}, \quad (\text{A.2.15})$$

$$\begin{aligned} \vec{\nabla} \times \vec{A} = & \frac{\hat{r}}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \\ & + \frac{\hat{\theta}}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \\ & + \frac{\hat{\phi}}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right], \end{aligned} \quad (\text{A.2.16})$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}. \quad (\text{A.2.17})$$

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